Daniel Rakita

Curriculum Vitae — August 2026

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Research Mission

The high-level goal of my lab is to enable robots and learning systems to act and improve in real-time, directly within the dynamic, uncertain environments of the real world. We develop algorithms for fast optimization, planning, and control that allow systems to continuously update their behavior as they operate, tightly integrating perception, action, and learning so that robots can react quickly, gather the right information, and adapt their strategies on the fly. Our work is motivated by domains where real-time adaptability has meaningful societal impact: home and assistive robotics, where systems must operate safely alongside people; healthcare and robotic surgery, where precision and real-time feedback are essential; and disaster response, where robots must act under uncertainty with limited prior information.

Appointments

2022 - Current	**Assistant Professor** Yale University, Department of Computer Science
2015-2022	**Graduate Researcher** University of Wisconsin-Madison Visual Computing Lab and Human-Computer Interaction Lab
2018, 2019	**NREIP Researcher** Naval Research Lab, Washington, D.C.

Education

2015-2022	**Ph.D. in Computer Science** University of Wisconsin-Madison *Advisors: Michael Gleicher and Bilge Mutlu*
2015-2017	**Masters of Computer Science** University of Wisconsin-Madison
2013-2015	**Undergraduate work in computer science** University of Wisconsin-Madison
2008-2012	**Bachelor of Music in Performance** Indiana University-Bloomington Jacobs School of Music

Selected Honors Awards

2023	Best Paper Award Winner, *ACM/IEEE Conference on Human-Robot Interaction (HRI)*
2022	Outstanding Graduate-Student Research Award, *UW-Madison*
2021	Outstanding Reviewer Award, *Selected by IROS Conference Paper Review Board, Top 4 of 3,942*
2021	Cisco Graduate Student Fellowship Recipient, *UW-Madison*
2021	Three Minute Thesis Competition Finalist, *UW-Madison*
2020	Best Paper Award Finalist, *ACM/IEEE Conference on Human-Robot Interaction (HRI)*
2019	Microsoft PhD Fellowship Recipient
2018	Best Paper Award Winner, *ACM/IEEE Conference on Human-Robot Interaction (HRI)*, *Top 4 of 206 papers*
2017	NSF Graduate Research Fellowship Program Honorable Mention

2017 HRI Pioneer, *accepted to the selective workshop held at HRI 2017*

Selected Research Grants and Funding

2025 - 2026 **Advanced AI-Based Manipulation for USV Diesel Engine Maintenance** (Office of Naval Research)
Role: Key Investigator, Amount: $160,000
STTR Phase 1. Key Investigators: Brian Kononchik (Boston Engineering), Daniel Rakita (Yale), Brian Scassellati (Yale), John Steagall (Fairbanks Morse Defense).

2024 - 2027 **Robot Manipulation in Densely Cluttered Environments** (Office of Naval Research)
Role: Principal Investigator, Amount: $1,212,027
Co-PI(s): Brian Scassellati.

2021-2022 **Cisco Graduate Student Fellowship** (Cisco)
Amount: one year PhD tuition and stipend

2019-2020 **Microsoft PhD Fellowship** (Microsoft)
Amount: $84,000 for tuition, stipend, and travel funds

Selected Publications

Conference

2026 Sun, X., Wang, Y., Yang, S., Chen, Y., and **Rakita, D.** Hybrid Diffusion Policies with Projective Geometric Algebra for Efficient Robot Manipulation Learning. *ICRA*.[Link]
Oral Presentation

Liang, C., Sun, X., Wang, Q., and **Rakita, D.** Turning Stale Gradients into Stable Gradients: Coherent Coordinate Descent with Implicit Landscape Smoothing for Lightweight Zeroth-Order Optimization. *ICML*.

Wang, Q., Abdellall, O., Gao, T., Sun, X., and **Rakita, D.** Subsecond 3D Mesh Generation for Robot Manipulation. *ICRA*.[Link]

Liang, C. and **Rakita, D.** Robust and Efficient MuJoCo-based Model Predictive Control via Web of Affine Spaces Derivatives. *IROS*.[Link]

2025 Liang, C., Wang, P., Xu, A., and **Rakita, D.** ad-trait: A Fast and Flexible Automatic Differentiation Library in Rust. *IROS*.[Link]

Hoffmeister, L., Scassellati, B., and **Rakita, D.** Towards Zero-Knowledge Task Planning via a Language-based Approach. *IROS*.

Rakita, D., Liang, C., and Wang, Q. Coherence-based Approximate Derivatives via Web of Affine Spaces Optimization. *RSS*.[Link]

Sun, X., Yang, S., Chen, Y., Fan, F., Liang, Y., and **Rakita, D.** Dynamic Rank Adjustment in Diffusion Policies for Efficient and Flexible Training. *RSS*.[Link]

2024 Hoffmeister, L., Scassellati, B., and **Rakita, D.** Sequential Discrete Action Selectionvia Blocking Conditions and Resolutions. *IROS*.[Link]

2023 Schoen, A., Sullivan, D., Zhang, Z., **Rakita, D.**, and Mutlu, B. Lively: Enabling Multimodal, Lifelike, and Extensible Real-time Robot Motion. *HRI*.[Link]
Best Paper Award Winner

Patel, V., **Rakita, D.**, and Dollar, A. An Analysis of Unified Manipulation with Robot Arms and Dexterous Hands via Optimization-based Motion Synthesis. *ICRA*.[Link]

Wang, Y., Praveena, P., **Rakita, D.**, and Gleicher, M. RangedIK: An Optimization-based Robot Motion Generation Method for Ranged-Goal Tasks. *ICRA*.[Link]

2022 **Rakita, D.**, Mutlu, B., and Gleicher, M. PROXIMA: An Approach for Time or Accuracy Budgeted Collision Proximity Queries. *RSS*.[Link]

2020 Praveena, P., **Rakita, D.**, Mutlu, B., and Gleicher, M. Supporting Perception of Weight through Motion-induced Sensory Conflicts in Robot Teleoperation. *HRI*.
Best Paper Award Nominee

2018 **Rakita, D.**, Mutlu, B., and Gleicher, M. An Autonomous Dynamic Camera Method for Effective Remote Teleoperation. *HRI*.
Best Paper Award Winner

2016 Bodden, C., **Rakita, D.**, Mutlu, B., and Gleicher, M. Evaluating Intent-Expressive Robot Arm Motion. *RO-MAN*.
Best Paper Nominee

Journal

2026 Bader, J., Sun, X., Rosenfelt, T., Ramirez-Hardy, A., Sapir, N., Scheub, R., Vitchutripop, T., Khanna, A., Pantel, H., and **Rakita, D.** AI-Powered Semantic Segmentation Model for Enhanced Ureteral Mapping and Real-Time Instrument Feedback in Robotic Colorectal Surgery. *Journal of Robotic Surgery*.

Bader, J., Sun, X., Rosenfelt, T., Ramirez-Hardy, A., Vitchutripop, T., Pantel, H., Khanna, A., and **Rakita, D.** Deep learning approach for critical exposure during division of the inferior mesenteric artery in colorectal surgery. *Journal of Robotic Surgery*.[Link]

2021 Chamzas, C., Quintero-Pena, C., Kingston, Z., Orthey, A., **Rakita, D.**, Gleicher, M., Toussaint, M., and Kavraki, L. MotionBenchMaker: A Tool to Generate and Benchmark Motion Planning Datasets. *IEEE Robotics and Automation Letters*.[Link]

Rakita, D., Mutlu, B., and Gleicher, M. Single Query Path Planning using Sample Efficient Probability Informed Trees. *RA-L/ICRA*.

Selected Invited Talks

2025 *The Applied Planning, Learning, and Optimization Toolbox: A Flexible Rust-Based Software Suite for Robotics Research*.
Rust for Robotics Workshop, ICRA

2022 *Intuitive Robot Shared-Control Interfaces via Real-time Motion Planning and Optimization*.
Cornell University

2022 *Generating Accurate, Feasible, and Coordinated Bimanual Robot Motions in Real-time*.
Workshop on Bimanual Manipulation, ICRA

2021 *Methods and Applications for Generating Accurate and Feasible Robot-arm Motions in Real-time*.
KavrakiLab, Rice University

2021 *Methods and Applications for Generating Accurate and Feasible Robot-arm Motions in Real-time*.
Talking-Robotics Series

Selected Teaching

CPSC 4870/5870 3D Spatial Modeling and Computing, Yale University *course I designed*
Several areas of computer science and related fields must model and compute how objects are situated in three-dimensional space over time... This course will teach students how to computationally model the spatial configuration of and spatial relationships between objects over time. — Spring 2024. Course Rating: 4.5/5.0, Spring 2026. Course Rating: 4.6/5.0

CPSC 4850/5850 Applied Planning and Optimization, Yale University *course I designed*
This course introduces students to concepts, algorithms, and programming techniques pertaining to planning and optimization... — Spring 2023. Course Rating: 4.5/5.0, Fall 2023. Course Rating: 4.5/5.0, Fall 2025. Course Rating: 4.4/5.0

CPSC 685 Topics on Robot Motion Generation, Yale University *course I designed*
This course focuses on concepts, approaches, and algorithms related to robot motion generation. Students will read, summarize, present on, and discuss papers and textbook chapters related to search-based path planning, sampling-based path planning, inverse kinematics, and trajectory optimization. — Fall 2022. Course Rating: 4.8/5.0

CPSC 472 Intelligent Robotics, Yale University *Guest Lecture* — Fall 2022

CS/ Psych 770 Human-Computer Interaction, University of Wisconsin-Madison *Guest Lecture* — Spring 2020

Selected Advising

Ph.D. Students

2023 - current	Liam Merz Hoffmeister
	Xiatao Sun
2024 - current	Chen Liang
	TJ Vitchutripop
	Peter Wang
2025 - current	Roshan Klein-Seetharaman

Masters Students

2024 - 2025	Alan Li
	Yinliang Chen

Selected Service

2025-Current	**Associate Editor**, Transactions on Human-Robot Interaction (THRI)
2024	**Program Committee Member**, HRI
2024	**Associate Editor**, ICRA
2021	**Session Chair**, ICRA session Optimization-Based Motion Planning
2021-Current	**Review Editor**, Frontiers in Robotics and AI

Selected Media

Yale News — *Grasping the future with a robotic arm-hand combo*

Techcrunch — *This robot learns its two-handed moves from human dexterity*

Tech Xplore — *Shared control allows a robot to use two hands working together to complete tasks*

Cosmos, The Science of Everything — *Breaking: robot makes breakfast*

Milwaukee Journal Sentinel — *UW team designs robot hands that work together*

Technical Skills

Programming	Rust, Python, C++, C, C#, Java, OpenGL, ROS, MATLAB, JavaScript, HTML, CSS, WebGL
Software	Blender, 3dsMax, Unity, MotionBuilder, Photoshop, Illustrator, Premier Pro, After Effects, Maya, MudBox, Office